ML Network Intrusion Detector

Software Requirements and Setup Guide

1. System requirements

• Operating system: Windows 10/11, Linux, or macOS
• Python: 3.10–3.12 recommended
• RAM: 8 GB recommended; larger datasets may require more
• Storage: enough free space for the dataset, trained model, and generated outputs
• Input: labeled network-flow CSV with feature columns and a target/label column

2. Python packages

Package	Purpose	Version constraint
pandas	Read and manipulate CSV data	>=2.0, <3.0
NumPy	Numerical operations and missing/infinite-value handling	>=1.24, <3.0
scikit-learn	Preprocessing, Random Forest, train/test split, metrics	>=1.3, <2.0
joblib	Save and load the trained pipeline	>=1.3, <2.0

3. Installation

```
python -m venv .venv
# Activate the environment for your operating system
python -m pip install --upgrade pip
pip install -r requirements.txt
```

4. Run the project

```
python src/network_intrusion_detector.py train --data data/network_flows.csv --label-column Label
python src/network_intrusion_detector.py predict --data data/new_flows.csv
```

5. Dataset notes

The dataset is not bundled with this project. Obtain a suitable public labeled dataset (for example, CIC-IDS2017, CSE-CIC-IDS2018, UNSW-NB15, or NSL-KDD) from its official or authorized source. Review its license, citation requirements, label schema, and data quality. Update the command to match the actual CSV path and target column.

6. Outputs

Training writes a serialized pipeline to **artifacts/network_intrusion_model.joblib** and metrics to **artifacts/evaluation_metrics.json**. Prediction writes a CSV containing predicted classes and optional maximum class probabilities.

Security note: this is an educational offline classifier, not a live packet sensor or a production intrusion-prevention system. Do not use predictions to block traffic without independent validation and human oversight.